

VERSION HISTORY

Version	Date	Revision
1.0.0	April 30, 2026	Created

JUNO TH TILT IFM PAYLOAD DESCRIPTION EN - ZJF201

This chapter describes the structure of the telemetry data. The number of bytes in the payload depends on the sensor configuration. The data structure depends on which sensors are present in your product. In principle, every version has a header. This header contains information about the version and status of the sensor, as well as a master measurement value. This measurement value can simply be assumed to be the current measurement value. The header is followed by an additional payload section that contains further information, such as specific parameters for the measurement principle or settings for position and opening detection. Not every sensor has an additional payload section.

Terms	Description
Byte No.	Byte number starting at 1
Alias	Descriptive name of the variable
Description	Description of the variable
Label	Label in the data converter
Unit	Unit of the variable
Datahub .json key	Key in .json when using the Datahub

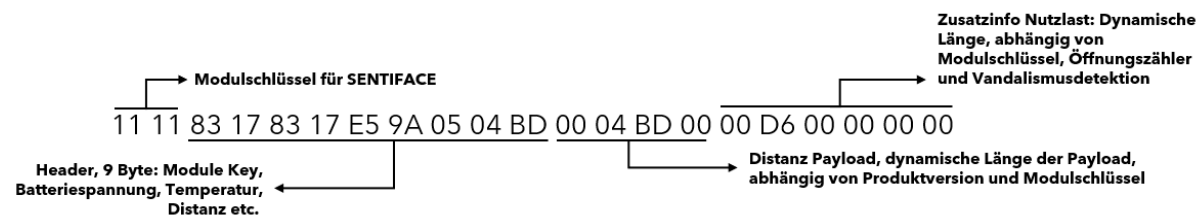
STRUCTURE OF THE MODULE KEY

Byte 1		Byte 2	
Bits 7-4	Bits 3-0	Bits 7-4	Bit 3 - 0
Base ID module, e.g., Sentiface, Senticom, Sentivisor	Major version (SW/HW version)	Minor version (SW/HW version)	Product version (sensors, e.g., TH, THL, ACC, ...)

The module key of the SENTIFACE module can be extracted from the first 2 bytes of each uplink. The module key is required for the downlink.

UPLINK EXAMPLE (NOT APPLICABLE TO JUNO)

This example describes how to read and interpret the payload:



Given is the following payload example for the Helios pressure sensor:

11 11 FE 1A D5 95 06 03 00 23 BE

Bytes	1	2	3	4	5	6	7	8	9	10	11
HEX	11	11	FE	1A	D5	95	06	03	00	23	BE
Description	Module key	Module Key	Uplink Counter	Battery Voltage	Battery Voltage	Temperature	Alarm Flag	Alarm Flag	Measurement Status	Pressure mbar	Pressure mbar

JUNO PAYLOAD DESCRIPTION

Byte Number	Alias Name	Description	Unit	Name in Payload Decoder	Module key	Datahub .json Key
1 - 2	Module key	Byte 1: 4-bit MSB: Module code. For Mioty, exclusively the "Sentiface" telemetry module, i.e., always 0x1. 4-bit LSB: Major version. Firmware of the same major version but a higher minor version remains compatible with older decoders. However, the number of bytes may change! Byte 2: 4-bit MSB: Minor version. New software versions may incrementally add new telemetry data to the data packet. 4 LSB bits: Sub/Product version. In Helios, these bits encode the specific configuration of the connected sensors.		module_key	XXX1	module_key
3	Uplink Counter	Uplink Counter: Protocol-independent uplink counter with a maximum value of 255. After reaching 255, the counter resets to 0.		uplink_counter	XXX1	uplink_counter
4 - 5	Battery voltage	Current battery voltage in millivolts //Voltage in volts battery_voltage: (dataBytes[3] << 8 dataBytes[4])/1000.0	mV	battery_voltage	XXX1	battery_voltage
6	Internal temperature	Sensor temperature: Internal_temperature = bytes[6] - 128; For this sensor, this value represents the controller's temperature reading and is an inaccurate temperature value of +/-2°C	°C	internal_temperature	XXX1	internal_temperature
7	Alarms	The alarms are coded as follows. Caution: Multiple alarms may be active at the same time!		humidity	XXX1	humidity

		DEC 1, BIN 0000 0001: High temperature limit exceeded DEC 2, BIN 0000 0010: Temperature has fallen below the low limit DEC 4, BIN 0000 0100: Temperature delta alarm DEC 8, BIN 0000 1000: Relative humidity high limit exceeded DEC 16, BIN 0001 0000: Relative humidity limit low not reached DEC 32, BIN 0010 0000: Relative humidity alarm delta				
8 - 9	Temperature	Exact temperature from the SHT40, as described in the datasheet. <code>decoded.temperature = ((bytes[idx++] << 8) bytes[idx++]) / 10 - 100;</code>	°C	temperature	XXX1	temperature
10	Relative humidity	Relative humidity from the SHT40, as described in the datasheet.	%RH	humidity	XXX1	humidity
11	Accelerometer status	0: Acceleration sensor without error - Other: Acceleration sensor error		acc_status	XXX1	acc_status
12	Sensor orientation	Displays the orientation of the sensor 0: Vertical: Lying on its side 1: Face Up 2: Face Down 3: Error		orientation	XXX1	orientation
13	Angle	Returns the angle in degrees, resolution 1°	°	angle	XXX1	angle
14	Open	Indicates whether the flap is open or closed		is_open	XXX1	is_open
15	Relative opening counter	Indicates how many openings have been performed since the last uplink		open_cnt_rel	XXX1	open_cnt_rel
16	Open counter lifetime	Indicates how many openings have been performed during the entire operation		open_cnt_total	XXX1	open_cnt_total